SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Fundamentals of Data Science **COURSE CODE: DSA04**

COURSE OUTCOME COVERED

CO1: *Analyze the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL2)*

QUESTIONS: 20 **Total Marks:100**

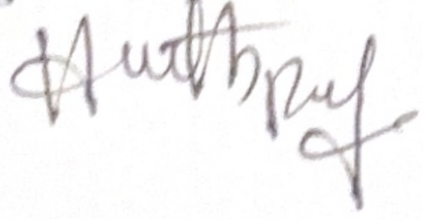
Annexure A

SHORT ANSWER TEST

Code of Conduct:

I N·HemanthReddy (192472215) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessement Tool - 1.

Name :- N. Hemanth Reddy

Reg NO :- 192472215

Course Code :- DSA 0401

Course Name :- fundamental of Data-Science.

Faculty Name :- Dr. P. Oma Rani.

N. Hemanth Reddy
192472215

DSA0401 - Fundamental of Data Science.

Assessment 7001 - 1.

1. A Hospital has patient records for many years. How can Data Science be applied?

A. Data Science Can analyse historical patient records to identify patients in diseas, treatments, and health Conditions. It can help hospitals predict diseases, improve treatment decisions, and manage healthcare resources effectively.

Example:- predicting which patients may develop diabets.

2. A Bank wants to detect Suspicious transactions. How Can Data Science be useful?

Data Science Can analyze transaction data to identify unusual patterns Such as unexpected locations, large amounts, or frequent transactions. Machine leasning algorithms Can detect these patterns and help identify fraudlent transactions.

Examples:- Detecting an unsual large payment from a Customer's account.

3. Q) School wants to improve Student performance. Mention One benefit of Data Science in this case.

A. Data Science Can analyze Students marks, attendance, assignments, and learning patterns. It Can identify Students who are performing poorly and help teachers provide personalized Support.

Example :- Identifying Students who need extra Classes.

4. Q) transport department wants to reduce traffic congestion. How Can Data Science be applied?

A. Data Science Can analyze traffic data Such as vehicle Counts, trave times, locations, and peak hours. By identifying traffic patterns, the department Can Optimize traffic Signals and Suggest alternative routes.

Example :- finding roads with heavy traffic during Peak hours.

5. Q) Company has data in excel tables, Customer reviews, images, and videos. which facet of data is Shown here?

A. This Shows the vasiety facet of Data. Data is available in differement forms Such as Structured data.

(Excel tables) and unstructed data (reviews, images and videos.

Example :- Excel data, Customer reviews, photos and videos.

6. A univerty Stores Student details in rows and columns. what type of data Structure is this?

A. This is Structured Data. Structed data is Organized in a fixed format using rows and columns. Each row represents a record, while each column represents a field or attribute.

7. A company collects a very large amount of daily user activity data forms its Mobile app. which Big Data characterstic is applied?

A. The Big Data Characterstic is volume. Volume refers to the very large amount of data generated, collected, and Stored by organizations. Large amount volumes require efficient Storage and processing technologies.

Example :- Millions of user activities generated every data.

8. A Business wants to convert Customer data into useful Sales decisions. which Big Data characteristic is important?

A. The important Big Data Characteristic is value. value refers to extracting useful information and insights from data to Support bussiness decisions. Customer Data can be analysed to understand and improve Sales Strategies.

Example :- Finding which products customers buy the most.

9. Q) A Company wants to process large datasets using parallel computation. which Big Data tool can be used?

A. Apache Spark Can be used to process large datasets using parallel Computation. It divides data-processing tasks across multiple Computers, allowing large amounts of data to be processed quickly.

Example :- Processing Millions of customer transactions quickly.

10. A Company wants to visualize Big Data results for decision-makers. which tool or method Can be applied?

A. Data visualization Can present Big Data results through charts, graphs, dashboards, and reports. Tool Such as Tableau, power BI, and python help decision-makers understand patterns and make data-driven decisions.

Example :- Showing Monthly Sales using a bar Chart.